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17 September 1971
Superseding
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13 March 1970

MILITARY SPECIFICATION

Test Set, Programmer AN/USM-299A

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The test set covered by this specification shall be for Intermediate Level use and shall be capable of testing the vunctional operation and accuracy of the Programmer Group, Missile Acquisition AN/ASA-63, or AN/ASA-63A. To accomplish this, the Test Set, Programmer, AN/USM-299A, hereinafter referred to as the Test Set, contains the circuits necessary to supply primary power, functional control, simulated data inputs, and readout devices for the programmer circuits.

1.2 Classification - The Test Set shall consist of the following items:

Item	Type Designation	Appl. Para.
Test Set Subassembly	TS-2709A/USM-299	3.5.1
Case, Test Set	CY-6503/USM-299	3.5.2
Cover, Test Set	CW-995/USM-299	3.5.2.2
Wiring Harness, Branched	W1 of AN/USM-299	3.5.3
Cable Assembly, Radio Frequency	W2, W3, and W4 of AN/USM-299	3.5.4
Wiring Harness, Branched	W5 of AN/USM-299	3.5.5

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6718	Control Panel, Rack Mounted
MIL-M-7793	Meter, Time Totalizing

FSC- 6625

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MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures: Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-F-19207	Fuseholder, Extractor Post Type, Blown Fuse Indicating and Nonindicating
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for use with Electronic and Fire Control System, Gen. Spec. for
MIL-S-23603	System Readiness/Maintainability. Avionics System Design Gen. Spec. for
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Test, Exponential Distribution

DRAWINGS

Military

MS24264	Connectors, Receptacles, Electrical - Flange Mount, Miniature
MS27035	Connector Receptacle, Electrical, Series BNC, UG-625B/U
MS21993	Cable Assemblies, Power Electrical

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2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Parts and materials requirements shall conform to MIL-T-21200.

(b) Nonrepairable subassemblies, as outlined in MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the mount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

(c) When previously produced models of the Test Set did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall be described by a contractor procurement specification and must be approved by the procuring agency. Microelectronic Modular Assemblies (Microelectronic Circuits) (MICS) shall be contained within Maintenance Modules, (for example, on printed circuit cards). The choice of Microelectronic Modular Assemblies shall be based on at least two sources of supply, which can be used interchangeably.

3.2.2 Modules, Maintenance - The electronic portions of the Test Set shall be divided into maintenance modules (see MIL-S-23603). Maintenance modules shall normally be considered repairable.

3.3 Design and Construction - The Test Set shall conform with all the applicable requirements of MIL-T-21200 for design, construction and workmanship except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the Test Set including cables, shall be a minimum consistent with good design and shall not exceed 55 pounds.

3.3.2 Reliability

3.3.2.1 Operational Stability - The Test Set shall operate with satisfactory performance, continuously or intermittently for a period of at least 24 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

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3.3.2.2 Operating Life - The Test Set shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the Test Set, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The Test Set shall be tested in accordance with the requirements of MIL-STD-781 paragraph 4.1.1, for a period of 12 hours. The test results shall indicate a MTBF of not less than two hundred (200) hours.

3.3.2.4 Time Totalizing Meter - The Test Set shall contain time totalizing meters in accordance with MIL-M-7793. Meter shall be included in the following units.

TS-2709A/USM-299 MS17322-6

3.3.3 Cables and Connectors - The Test Set shall provide for the use of cables and connectors in accordance with MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel, and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.3.3.1 Cable Assemblies, Power Electrical - Applicable input power cable and connections shall conform to MS21993 except the connectors for connections to Test Set.

3.3.4 Interchangeability - The Test Set shall meet the interchangeability requirements as defined in MIL-T-21200.

3.3.5 Interference Control - The generation of radio interference by the Test Set and the vulnerability of the Test Set to radio interference shall be controlled within the limits of MIL-I-6181.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room Ambient (30±10°C) (86±18°F)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient
Input Power Voltage	115±1.0 VAC, Three Phase, 400±5 CPS 27±2 VDC

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3.3.9 Service Conditions - The Test Set shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for class 3 equipment.

3.3.10 Warm-up Time - The time required for the Test Set to warm-up prior to operations shall be kept to a minimum and shall not exceed 30 minutes under standard conditions and 45 minutes at extreme service conditions.

3.3.11 Primary Input Power Requirements - The Test Set shall be capable of satisfactory operation when supplied input power having characteristics as defined in MIL-T-21200. The power required shall be of the following types and shall not exceed the specified amounts.

(a)	AC Power (Three Phase)	
	Operating Voltage Limits	103.5 to 126.5 volts
	Operating Frequency Limits	380 to 420 cps
	Power (3ϕ)	20 VA (max)
(b)	DC Power	
	Operating Voltage Limits	25 to 29 volts
	Current	1 Amp (max)

3.3.11.1 Voltage Protection - The Test Set shall not be damaged when exposed to transient AC voltages as defined in Figure 2 of MIL-STD-704, and shall automatically resume normal operation when the voltage returns within limits of normal operation.

3.3.11.2 Fuses and Fuse Holders (Panel-Mounting Type) - Fuses shall conform to MIL-F-19207. The fuse holder legends shall identify each supply fused and the corresponding fuse rating.

3.4 Performance - The performance of the Test Set shall be set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

3.5 Detail Requirements

3.5.1 Test Set Subassembly TS-2709A/USM-299

3.5.1.1 Function - The Test Set subassembly, in conjunction with supplied equipment, shall be capable of testing the operation and accuracy of the Programmer Group, Missile Acquisition, AN/ASA-63 or AN/ASA-63A.

3.5.1.2 Form Factor-The Test Set subassembly TS-2709A/ USM-299 shall be housed in the Case, Test Set CY-6503/USM-299 which is a combination case and fulfills the requirements of MIL-T-21200. The overall dimensions of the Test Set with cover installed shall not exceed 22 inches by 16 inches by 15 inches. When removed from the combination case, the subassembly shall be of such a form factor that it can be mounted in a standard 19-inch rack per MIL-C-6718B.

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3.5.1.3 Weight - The weight of the Test Set subassembly shall not exceed 25 pounds.

3.5.1.4 Content of Unit - The Test Set subassembly shall contain the following assemblies, subassemblies and circuits:

(a) 1500 CPS Oscillator - The Test Set shall contain a 1500 oscillator to simulate the missle audio signal. The 1500 CPS signal amplitude shall be controlled by the front panel AUDIO control knob and shall be adjustable from 0.5VRMS or less to 3.0 VRMS or more.

(b) 125 CPS Oscillator - The Test Set shall provide as an output to the Programmer under test a 125 ± 1 CPS signal to represent the missile reference signal. The 125 CPS output shall be fixed and have an amplitude of 1.6 ± 0.5 VRMS with a maximum harmonic distortion of 2.5 percent.

(c) Phase Meter Assembly - The Test Set shall provide the means to measure the phase angle of a 125 ± 1 CPS signal at various test points in the Programmer unit under test with respect to the reference signal described in 3.5.1.4(b) which is defined as having a zero phase angle. The phase angle of the signal under test shall be read directly on the front panel PHASE METER. The phase meter shall measure angles from 0° to 360° and shall read positive for a lagging phase shift. The phase measurement accuracy shall be ± 0.75 degrees for signals with amplitudes from 1.0 to 2.5 volts (RMS) and a maximum harmonic distortion of 2.5 percent.

(d) Radar Simulator - The Test Set shall contain two 400 CPS synchros which simulate radar azimuth and elevation antenna angles. The design shall be such that either (1) both the azimuth signal and elevation signal are variable from front panel controls, or (2) one synchro is variable and one synchro is locked at 0 degrees such that the elevation and azimuth signals can be switched between the variable synchros and locked synchro by the front panel RADAR INPUT SIMULATOR control. Both synchros shall provide 11.8 VAC ± 5%, 400 CPS, 3 wire signals at a scale factor of 206 mv ± 5%, per degree over a range of ± 45 degrees. The phasing shall be such that with the S2 voltage in phase with the 400 CPS supply Phase C, a zero degree antenna angle is simulated. With S3 to S1 nulled an UP signal for the Elevation output or a RIGHT signal for the Azimuth output shall be provided when the front panel controls are rotated from zero degrees toward the +45 degree dial calibration. A zero degree to 360 degree dial calibration is acceptable.

(e) Attack Angle - The Test Set shall provide a DC voltage which represents the aircraft attack angle. At least two DC voltages shall be provided between 0 and +30 VDC.

(f) Logic Test Circuitry - Circuitry shall be provided to simulate the test logic of the Programmer.

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(g) Acquisition Lambda Amplifier - The Test Set shall provide a 125 Hz Acquisition Lambda signal variable in phase from 0° thru 360° relative to the 125 Hz defined in paragraph 3.5.1.4 (b). Its amplitude shall be 2.75 ± 0.02 VRMS into a load of 22 kilohms.

3.5.1.5 Controls - The following controls and indicators, labeled as specified, shall be provided on the front panel of the Test Set.

(a) POWER ON - An illuminated push-type main power switch shall be provided on the front panel to connect or disconnect the incoming primary power. The switch shall be illuminated when power is "ON".

(b) SCAN-SLAVE - An illuminated push-type switch shall be provided on the front panel to place the Test Set in the correct operating condition during Logic Test.

(c) SCOPE SELECT - A switch shall be provided to switch scope inputs from those normally programmed to display phase null error, also selects signal for fine alignment of phase adjustments.

(d) LEAD ORDER - L/O ORDER - L/O INDICATE - Panel lamps shall be provided to indicate when lead order, lock-on order and lock-on indicate signals are provided by the unit under test. The indicator shall also be a momentary panel switch, which when depressed interrupts the Radar lock-on order signal.

(e) PROGRAM ORDER - Panel lamps shall be provided to indicate the program order from the Programmer under test.

(f) PHASE A, B, C, AND +28 VDC LAMP - A fuse holder with an illuminated legend shall be provided for each of the main power lines to indicate when a fuse is blown.

(g) VOLTS DC - A DC voltmeter shall be provided.

(h) VOLTS AC - An AC meter shall be provided.

(i) PHASE NULL - A zero to 3 volt rectifier type meter shall be provided as a part of the phase metering circuit.

(j) PHASE METER - A phase shifter with a calibrated dial for reading phase angle shall be provided as a part of the phase metering circuit to indicate the measured phase difference between reference signal and measured signal.

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(k) RADAR INPUT SIMULATOR - A minimum of one control with a dial calibrated in degrees shall be furnished to control elevation and azimuth synchro signals.

(l) LAMP TEST - A pushbutton switch shall be provided to test the operation of the lamps circuits on the front panel.

(m) PILOTS L/O - A pushbutton switch shall be provided to simulate the Pilots lock-on order to the Programmer being tested.

(n) AC VOLTS - A four position rotary switch shall be provided to select one of four possible ranges in which to use the VOLTS AC meter.

(o) RADAR INPUT - A two position rotary switch shall be provided to select either variable azimuth information or variable elevation information as the simulated radar inputs to the Programmer.

(p) PWR SUPPLY TEST - A six position rotary switch shall be provided to select the various outputs of the Programmer power supplies for measurements.

(q) ATTACK ANGLE - A three position rotary switch shall be provided to select one of two DC voltages to represent the attack angle of the aircraft and to monitor the output of the unit being tested on the VOLTS DC meter under each condition.

(r) LOGIC TEST - A twelve position rotary switch shall be provided to select the proper logic inputs to the Programmer to satisfactorily test the logic circuity.

(s) **MSL SELECT - A four-position rotary switch shall be provided to select the proper Station Identification and Missile Identification voltages for each of four desired missiles.**

(t) MSL SIMULATOR - A two-position rotary switch shall be provided to select outputs of a particular missile or internal simulated missile outputs.

(u) SYSTEM ALIGN - A maximum of four rotary switches shall be provided to select the test points necessary to monitor the proper operation of the Programmer under test.

(v) AUDIO CONTROL - A control shall be provided to adjust the amplitude of the 1500 CPS signal.

(w) TIME TOTALIZER - A time totalizer indicator shall be provided on the front panel to measure the running time of the Test Set.

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3.5.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
70J1	MS24264R18B31SN	Signal Inputs and Outputs
70J2	MS24264R18B31S6	Signal Inputs and Outputs
70J3	MS24264R16B24PN	Signal Inputs and Outputs
70J4	MS24264R16B24S6	Missile Inputs and Outputs
70J5	MS24264R14B7PN	Power Inputs
70J6	MS27035-625B	Signal Outputs to Channel A of Oscilloscope
70J7	MS27035-625B	Signal Output to Channel B of Oscilloscope
70J8	MS27035-625B	Synch Output

3.5.2 Case, Test Set CY-6503/USM-299. The Test Set shall be provided with a combination case which shall conform to MIL-T-21200.

3.5.2.1 Size - The overall dimensions of the Case, Test Set shall be 22 inches by 16 inches by 15 inches

3.5.2.2 Cover, Test Set CW-995/USM-299 - The combination case shall be supplied with a detachable cover in accordance with MIL-T-21200.

3.5.2.2.1 Stowage - The cover shall contain provisions for stowing an instruction book, the cable assemblies, allen wrenches, extender cards, and spare fuses.

3.5.2.2.2 Size - The overall dimensions of the cover, Test Set CW-995/USM-299 shall be 22 inches by 4 inches by 15 inches.

3.5.2.3 Material - The case and front cover shall be constructed of aluminum. When approved by the procuring activity, other materials may be used, provided a savings in weight without a reduction in strength can be made.

3.5.2.4 Finish - The case shall be finished with gray enamel conforming to MIL-T-21200. The panel shall be finished with black synthetic lustreless enamel in accordance with TT-E-527.

3.5.3 Wiring Harness, Branched W1 of AN/USM-299 - The Test Set shall be furnished with a 6-foot power cable to provide power connections to the Test Set. One end shall terminate with a connector for connections to the Test Set. The branched ends of the cable shall be terminated in accordance with MS21993.

3.5.4 Cable Assembly, Radio Frequency W2, W3 and W4 of AN/USM-299 - The Test Set shall be furnished with three 6-foot RF cables to provide signal connections to an oscilloscope.

3.5.5 Wiring Harness, Branched W5 of AN/USM-299 - The Test Set shall be furnished with one 6-foot cable to provide connections between the Test Set and the Programmer under test. Each end of the cable shall be furnished with three connectors.

3.5.6 Extender Cards - Extender cards shall be provided in the Test Set to facilitate testing of the modules of the Programmer.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the Specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction Tests
(b) Initial Production Tests
(c) Acceptance Tests
(d) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on a Test Set representative of the production Test Set to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the Test Set meets all the requirements of this specification and the contract. Preproduction tests shall include environmental and interference tests in accordance with the requirements of MIL-T-5422 to the limits specified in MIL-T-21200. Unless specified by contract action Fungus, Sand, Dust, Explosion and Crash Safety tests shall not be required.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production Test Set shall be delivered prior to the approval of the preproduction sample. Prefabrication of production Test Set prior to the approval of the preproduction sample is at the contactor's own risk. The approval preproduction sample shall be retained by the contractor for his use in the fabrication and testing of Test Sets to be submitted for acceptance. The preproduction sample shall not be considered as one of the Test Sets under the contract.

4.2.4 Production Test Sets - Test Sets supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each Test Set shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection of production Test Sets and for Test Sets already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

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4.3 Initial Production Tests - One of the first ten production Test Sets shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This Test Set shall be selected by the procuring activity after the Test Set has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This Test Set may be subjected to any and all tests the procuring activity deems necessary to assure that the production test set is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete Test Set submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the Test Set.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of Test Set under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on Test Sets previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the Test Set shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the Test Set for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests
(b) Sampling Tests
(c) Reliability Assurance Tests
(d) Special Tests

4.4.1 Individual Tests - Each Test Set submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each Test Set accepted shall have passed the following tests:

(a) Examination of Product
(b) Operational Test
(c) Manufacturing Run-in Test

4.4.1.1 Examination of Product - Each Test Set shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each Test Set shall be operated long enough to permit the Test Set temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test - Each Test Set shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the Test Set. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the Test Set.

Temperature	Room ambient
Humidity	Room ambient
Vibration	Any selected frequency within the range of 20 to 30 CPS (excluding resonant points) and a minimum amplitude of±3 g's

The Test Set shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the Test Set shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the Test Set in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the Test Set shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future test sets. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - The Test Set selected for sampling tests shall first have passed the individual tests. Test Sets shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Test Set Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each Test Set selected for sampling tests shall be subjected to the following tests:

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(a) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(b) Operational test at certain environmental conditions. The conditions may vary for each Test Set tested and should be based on results of the preproduction, initial production, individual and special tests.

(c) Manufacturing run-in test specified in 4.4.1.3, except that the test duration shall be 120 hours with no restriction on the number of failures. However each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future Test Sets.

4.4.3 Reliability Assurance Tests - Unless specified by contract action, Reliability Assurance Tests shall not be required.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of Test Sets for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The Test Sets selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of Test Sets for special tests shall be made as follows:

(a) On an early Test Set after an engineering or material change.

(b) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Test Set Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(d) Where practical, include a test in the individual test to check all Test Sets for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on Test Sets that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Test Sets which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of test sets equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Test Sets Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Room ambient ($30^{o}\pm10^{o}$C)
Altitude	Normal Ground (0 - 5000 ft.)
Humidity	Room Ambient
Input Voltage	115±10% (at lowest applicable frequency) 27.0±2.0 volts DC
Vibration	None

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the Test Set temperature to stabilize. Periodically, the Test Set shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the Test Set, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(a) Part type number

(b) The circuit reference symbol number

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(c) The part function

(d) Name of the manufacturer

(e) Nature of the failure

(f) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - A test set which has been subjected to the life test shall be reconditioned as follows:

(a) On completion of the life test, the Test Set shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the Test Set for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Test Sets - The Test Set which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the Test Set for acceptance.

4.8 Presubmission Testing - No item, part or complete Test Set shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - The Test Set which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the Test Set shall be preserved packaged packed and marked for the level of shipment specified in the contract or order, in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - The Test Set, Programmer AN/USM-299A is used to check the operation and alignment of the Programmer.

6.2 Performance Characteristics - Normal and limiting values of performance data shall be determined at input voltages of 27.5±0.5 VDC and 115±1.0 VAC 400 cps, 3-phase as applicable. These data are to be used in testing the Test Set at installation points, for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - *Minimum size and weight, simplicity of operation,* ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of the Test Set. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes *other than those specified in MIL-T-21200, it is desired their use be investigated.* When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule, non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With a few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent *specified.*

6.6 Notation of Revisions - A bar in the margin denotes a change made in that paragraph from the previous version of the document. This has been done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as

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written, irrespective of the marginal notations and relationship to the last previous issue.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

(a)	Oscilloscope	AN/USM-105A (or equivalent)
(b)	Multimeter	AN/PSM-4 (or equivalent)
(c)	VTVM	TS-505T/U (or equivalent)
(d)	Voltmeter	John Fluke Model 873A (or equivalent)
(e)	Wheatstone Bridge	Leeds-Northrup Model 5305 (or equivalent)

Project No. 6625-N512

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Naval Air Systems Command
Washington, D.C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-81760A(AS) TEST SET, PROGRAMMER AN/USM-299A

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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